1. **Part 1:**
   Carry out an initial exploration of the data and comment briefly on your findings.

- Outcome variable (quality) of the dataset is numeric. The dataset has multiple continuous predictors (acidity, alcohol, sulphates, etc.)

- The dataset has 1599 rows and 12 columns (dim(wine)).

- ‘str(wine)’ revealed that all variables are numeric.

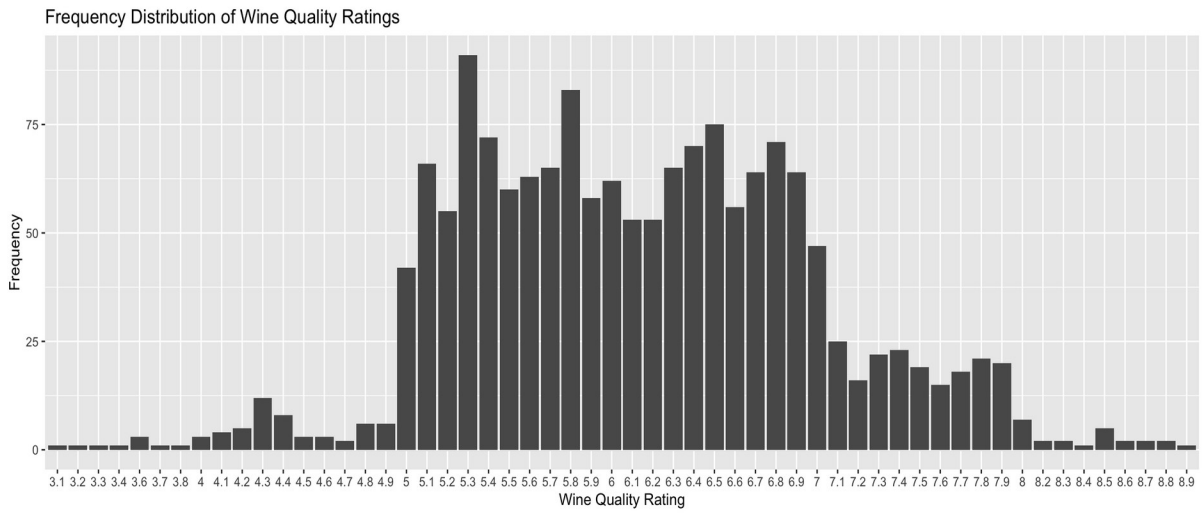
- ‘summary(wine)’ shows numeric columns have reasonable ranges. 'Min' value of 0 for Citric shows that some wines don’t have any citric.
The quality ratings range from 3.1 to 8.9, with a mean of 6.132, suggesting that most wines are of moderate quality, with fewer samples at the extremes. The values are concentrated between 5.5 and 6.7.
Below is the summary table for the wine dataset.

```
> summary(wine)
   fixacid        volacid          citric       residsugar      chlorides         freeSO2
 Min.   : 4.60   Min.   :0.1200   Min.   :0.000   Min.   : 0.900   Min.   :0.01200   Min.   : 1.00
 1st Qu.: 7.10   1st Qu.:0.3900   1st Qu.:0.090   1st Qu.: 1.900   1st Qu.:0.07000   1st Qu.: 7.00
 Median : 7.90   Median :0.5200   Median :0.260   Median : 2.200   Median :0.07900   Median :14.00
 Mean   : 8.32   Mean   :0.5278   Mean   :0.271   Mean   : 2.539   Mean   :0.08747   Mean   :15.87
 3rd Qu.: 9.20   3rd Qu.:0.6400   3rd Qu.:0.420   3rd Qu.: 2.600   3rd Qu.:0.09000   3rd Qu.:21.00
 Max.   :15.90   Max.   :1.5800   Max.   :1.000   Max.   :15.500   Max.   :0.61100   Max.   :72.00
   totalSO2        density             pH          sulphates         alcohol        quality
 Min.   :  6.00   Min.   :0.9901   Min.   :2.740   Min.   :0.3300   Min.   : 8.40   Min.   :3.100
 1st Qu.: 22.00   1st Qu.:0.9956   1st Qu.:3.210   1st Qu.:0.5500   1st Qu.: 9.50   1st Qu.:5.500
 Median : 38.00   Median :0.9968   Median :3.310   Median :0.6200   Median :10.20   Median :6.100
 Mean   : 46.47   Mean   :0.9967   Mean   :3.311   Mean   :0.6581   Mean   :10.42   Mean   :6.132
 3rd Qu.: 62.00   3rd Qu.:0.9978   3rd Qu.:3.400   3rd Qu.:0.7300   3rd Qu.:11.10   3rd Qu.:6.700
 Max.   :289.00   Max.   :1.0037   Max.   :4.010   Max.   :2.0000   Max.   :14.90   Max.   :8.900
```

Most of the variables have right-skewed distributions (Mean>Median).

- 'head(wine)' and ‘tail(wine)’confirms the first and last few records.

- The graph below illustrates the frequency distribution of wine quality ratings.



The distribution is unimodal, with many wines receiving ratings between 5 and 7, which indicates that most samples are of medium quality. Very low ratings (below 4) and very high ratings (above 8) are relatively uncommon. The slight right-skew of the distribution indicates that the high-quality wines are less common than average-quality wines.

Part 2:

Run a suitable regression model on the data and carry out a full regression analysis to achieve your best model, i.e. the simplest model that provides reliable predictions.

The outcome variable - wine quality is numeric and ordered. We have 11 independent variables, which are laboratory-based measurements. The wine quality is influenced by several factors simultaneously. So, the multiple linear regression model is the most suitable here.

The complete model has 11 explanatory variables.

Model 1

$y = \beta_0 + \beta_1 * x_1 + \beta_2 * x_2 + \beta_3 * x_3 + \beta_4 * x_4 + \beta_5 * x_5 + \beta_6 * x_6 + \beta_7 * x_7 + \beta_8 * x_8 + \beta_9 * x_9 +$

$\beta_{10} * x_{10} + \beta_{11} * x_{11}$

Where

x_1 = fixacid, x_2 = volacid, x_3 = citric, x_4 = residsugar, x_5 = chlorides, x_6 = freeSO2, x_7 = totalSO2, x_8 = density, x_9 = pH, x_{10} = sulphates, x_{11} = alcohol

```
Call:
lm(formula = wine$quality ~ wine$fixacid + wine$volacid + wine$citric +
    wine$residsugar + wine$chlorides + wine$freeSO2 + wine$totalSO2 +
    wine$density + wine$pH + wine$sulphates + wine$alcohol)

Residuals:
     Min       1Q   Median       3Q      Max
-2.57661 -0.44385 -0.01441  0.47731  2.18586

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     1.864e+01  2.319e+01   0.804 0.421496
wine$fixacid    1.730e-02  2.839e-02   0.609 0.542363
wine$volacid   -1.065e+00  1.325e-01  -8.036 1.80e-15 ***
wine$citric    -2.371e-01  1.610e-01  -1.473 0.140967
wine$residsugar 4.605e-03  1.641e-02   0.281 0.779053
wine$chlorides -2.090e+00  4.587e-01  -4.558 5.57e-06 ***
wine$freeSO2    3.713e-03  2.375e-03   1.563 0.118186
wine$totalSO2  -2.947e-03  7.972e-04  -3.697 0.000226 ***
wine$density   -1.385e+01  2.367e+01  -0.585 0.558358
wine$pH        -4.793e-01  2.096e-01  -2.287 0.022347 *
wine$sulphates  9.940e-01  1.251e-01   7.947 3.60e-15 ***
wine$alcohol    2.843e-01  2.897e-02   9.811  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.7089 on 1587 degrees of freedom
Multiple R-squared:  0.322,     Adjusted R-squared:  0.3173
F-statistic: 68.51 on 11 and 1587 DF,  p-value: < 2.2e-16
```

The estimated regression equation is :

$\hat{y}_i = 18.64 + 0.0173x_1 - 1.065x_2 - 0.2371x_3 + 0.0046x_4 - 2.09x_5 + 0.0037x_6 - 0.0029x_7 - 13.85x_8 - 0.4793x_9 + 0.994x_{10} + 0.2843x_{11}$

In Model 1,

p-value of fixacid = 0.5423, of residsugar = 0.779, of density = 0.5584.

They don't have a statistically significant effect on wine quality, so we are removing these 3 variables in Model 2.

Model 2

$y = \beta_0 + \beta_2 * x_2 + \beta_3 * x_3 + \beta_5 * x_5 + \beta_6 * x_6 + \beta_7 * x_7 + \beta_9 * x_9 + \beta_{10} * x_{10} + \beta_{11} * x_{11}$

```
Call:
lm(formula = wine$quality ~ wine$volacid + wine$citric + wine$chlorides +
    wine$freeSO2 + wine$totalSO2 + wine$pH + wine$sulphates +
    wine$alcohol)

Residuals:
     Min       1Q   Median       3Q      Max
-2.57116 -0.44140 -0.01327  0.48299  2.19129

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     5.1805276  0.5039606  10.280  < 2e-16 ***
wine$volacid   -1.0674808  0.1267841  -8.420  < 2e-16 ***
wine$citric    -0.2178055  0.1331655  -1.636    0.102
wine$chlorides -2.1453695  0.4403692  -4.872 1.22e-06 ***
wine$freeSO2    0.0038562  0.0023471   1.643    0.101
wine$totalSO2  -0.0030412  0.0007605  -3.999 6.65e-05 ***
wine$pH        -0.5712678  0.1455920  -3.924 9.09e-05 ***
wine$sulphates  0.9784184  0.1205244   8.118 9.40e-16 ***
wine$alcohol    0.2963921  0.0187307  15.824  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7083 on 1590 degrees of freedom
Multiple R-squared:  0.3218,    Adjusted R-squared:  0.3184
F-statistic:  94.3 on 8 and 1590 DF,  p-value: < 2.2e-16

>
```

The estimated regression equation is :

$\hat{y}_i = 5.18 - 1.0674x_2 - 0.2178x_3 - 2.1453x_5 + 0.0038x_6 - 0.003x_7 - 0.571x_9 + 0.9784x_{10} + 0.2963x_{11}$

In Model 2,

p-value of citric = 0.102 and of freeSO2 = 0.101.

They don't have a statistically significant effect on wine quality, so we are removing these 2 variables in Model 3.

Model 3

$y = \beta_0 + \beta_2 * x_2 + \beta_5 * x_5 + \beta_7 * x_7 + \beta_9 * x_9 + \beta_{10} * x_{10} + \beta_{11} * x_{11}$

```
Call:
lm(formula = wine$quality ~ wine$volacid + wine$chlorides + wine$totalSO2 +
    wine$pH + wine$sulphates + wine$alcohol)

Residuals:
     Min       1Q   Median       3Q      Max
-2.60316 -0.45264 -0.00773  0.46988  2.16572

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     4.6642274  0.4368644  10.677  < 2e-16 ***
wine$volacid   -0.9871518  0.1098032  -8.990  < 2e-16 ***
wine$chlorides -2.2469377  0.4352411  -5.163 2.74e-07 ***
wine$totalSO2  -0.0022773  0.0005537  -4.113 4.11e-05 ***
wine$pH        -0.4184325  0.1268703  -3.298 0.000995 ***
wine$sulphates  0.9688850  0.1203157   8.053 1.57e-15 ***
wine$alcohol    0.2915662  0.0183803  15.863  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.7093 on 1592 degrees of freedom
Multiple R-squared:  0.3191,    Adjusted R-squared:  0.3166
F-statistic: 124.4 on 6 and 1592 DF,  p-value: < 2.2e-16

>
```

The estimated regression equation is :

$$\hat{y}_i = 4.6642 - 0.9871x_2 - 2.2469x_5 - 0.0023x_7 - 0.4184x_9 + 0.9689x_{10} + 0.2916x_{11}$$

I chose Model 3 because it is the simplest model that provides reliable predictions. Even though the adjusted R-squared of Model 3 is almost the same as that of the previous model, Model 3 has only significant variables. This is the reason Model 3 is the most efficient and accurate model for the projection of wine quality.

Part 3:
Comment on the validity and predictive ability of your model.

In Model 1, the predictive model is linear, with the quality of the wine being measured using a weighted sum of the following eleven physicochemical variables: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free SO2, total SO2, density, pH, sulphates, and alcohol. All these variables are valid indicators of quality for the following reasons: they are established indicators of quality employed by experts, and they are widely cited in literature.

As for the appropriateness of the model from the statistical perspective, it will be appropriate if the major requirements for linear regression are satisfied. These include linearity between the predictors and the wine quality, independent sample observations, constant variance of the residuals, and normal errors. As much as it can be useful in improving the accuracy of wine quality predictions, the presence of many predictor variables can pose a problem of

multicollinearity, particularly among the sets of wine attributes like acidity, pH, and density. It should be assessed for statistical detection.

Once the optimal regression equation has been derived, it can be used to predict the outcome, y given set values of the independent variables.

$\hat{y}_i = 4.6642 - 0.9871x_2 - 2.2469x_5 - 0.0023x_7 - 0.4184x_9 + 0.9689x_{10} + 0.2916x_{11}$

Predicted y and Residuals have been calculated for the first 10 entries.

i	quality	volacids	chlorides	totalSO2	pH	sulphates	alcohol	Predicted y	Residuals
1	**3.4**	**0.58**	**0.074**	**47**	**3.25**	**0.57**	**9**	**5.63**	**-2.23**
2	**3.6**	**0.61**	**0.2**	**16**	**3.16**	**0.63**	**8.4**	**5.31**	**-1.71**
3	**3.1**	**1.185**	**0.097**	**14**	**3.63**	**0.54**	**10.7**	**5.37**	**-2.27**
4	**3.7**	**0.44**	**0.145**	**48**	**3.38**	**0.86**	**9.9**	**6.1**	**-2.4**
5	**4.0**	**1.02**	**0.084**	**11**	**3.48**	**0.49**	**11**	**5.67**	**-1.67**
6	**3.2**	**1.58**	**0.137**	**9**	**3.5**	**0.4**	**10.9**	**4.88**	**-1.68**
7	**3.3**	**0.815**	**0.267**	**29**	**3.32**	**0.51**	**9.8**	**5.16**	**-1.86**
8	**3.6**	**0.98**	**0.061**	**49**	**3.31**	**0.55**	**9.7**	**5.42**	**-1.82**
9	**3.6**	**0.875**	**0.082**	**14**	**3.4**	**0.52**	**10.2**	**5.64**	**-2.04**
10	**3.8**	**0.76**	**0.078**	**12**	**3.55**	**0.63**	**9.95**	**5.74**	**-1.94**